

This practice sheet is not meant for submission, and will not have any grade. You can ask me about any questions you have, including checking your answers.

Warning Even though this guide will not have any grade, it is important to avoid copying/plagiarism. Avoid the usage of LLMs (such as ChatGPT, DeepSeek, etc). When solving a graded assignment, this warnings will be actual rules that must be followed.

Context

This exercise guide is meant to practice what we learn during classes. This particular guide will cover the introduction to assembly and data representation lectures.

Exercises

Task 1 From the math functions in `standard.h`, implement the following ones:

SubTask 1 Function `max`.

SubTask 2 Function `is_odd`, `is_even`.

SubTask 3 Function `maximum` (**hard**).

Task 2 From the strings functions in `standard.h`, add and/or implement the following ones:

SubTask 1 Function `char_at` that given a string and an index, returns the character at that index (**new**).

SubTask 2 Function `length` (**hard**).

Task 3 From the boolean functions in `standard.h`, add and/or implement the following ones:

SubTask 1 Function `and`.

SubTask 2 Function `or`.

SubTask 3 Function `xor`.

SubTask 4 Function `not`.

SubTask 5 Function `all`, takes an array of `boolean` values, and a size, it returns `true` iff all elements are `true` (**hard**) (**new**).

Task 4 Using `objdump -d`, see how all `expression` related functions work with respect to arguments and value access.